

LY CHINH TAN

Biomedical Engineering | Physiological Time Series | Machine Learning

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[GitHub: LCT-dotcom](#) | [ResearchGate](#)

PROFILE

Fourth-year Biomedical Engineering student at HCMUT and first author of a published activity-recognition paper and a PPG manuscript under revision. Experience in wearable signal analysis, ECG domain adaptation and medical image processing, with an emphasis on reproducible experiments and subject-level evaluation. Currently developing embedded-systems knowledge through STM32 study.

EDUCATION

Ho Chi Minh City University of Technology, VNU-HCM

Engineering Physics, Biomedical Engineering major | August 2023–Present

English-taught program | Cumulative GPA: 3.1/4.0 | Semester 252 GPA: 3.7/4.0

RESEARCH

Human Behavior Recognition Using 2D Time-Series Data and Machine Learning Models

First author with Quang Linh Huynh | Journal of Physics: Conference Series 3180, 012004 (2026)

[DOI: 10.1088/1742-6596/3180/1/012004](#)

- Built an eight-class activity-recognition pipeline using 17-joint 2D pose sequences, torso normalization, 150-frame sliding windows, TSFEL and handcrafted kinematic/geometric features.
- Evaluated ensemble classifiers including HistGradientBoosting, XGBoost and LightGBM; analyzed subject-level performance and confusion between similar activities.
- Extended the accompanying repository with training-only feature selection, nested subject-level evaluation, Viterbi decoding and unseen-participant inference. Later five-subject LOSO results: 73.31% accuracy and 76.39% macro F1, separate from the published paper's original evaluation.

A Lightweight Temporal Model for Heart Rate Estimation from Wrist PPG under Daily-Life Motion

First author with Tuan Kiet Pham, Duy Trung Do, Phi Phong Pham and Quang Linh Huynh

BME11 manuscript submitted; under revision, decision pending | August 2025–Present

- Developed a PPG/accelerometer pipeline with synchronized windows, residual and spectral encoding, temporal attention, mixture-of-experts modeling and staged sequence inference.
- Ran strict 15-fold LOSO evaluation over 64,697 windows: raw macro MAE 4.307 ± 2.020 bpm; decoded MAE 4.278 ± 1.935 bpm. Reported the complete post-processing result separately (4.491 ± 1.896 bpm).
- Conducted six controlled ablation contrasts with paired subject-level tests and Holm correction. Residual CNN and temporal attention showed supported contributions; no embedded-device validation is claimed.

WORK EXPERIENCE

Coordinator | NCKU Overseas Hub in Vietnam

May 2026–Present

Intern | PS Medical

June–August 2026 | Approximately two months

Company specializing in laser equipment for aesthetic applications.

SELECTED PROJECTS

AD8232 ECG Domain Adaptation | Internship project

- Evaluated a three-class MIT-BIH/SVDB-trained ECG classifier on a 30-minute single-lead AD8232 recording; audited ventricular ectopic beat failures, channel representation and temporal-context mismatch.
- Adapted the Stage-B bidirectional GRU with attention while freezing Stage A and calibration rules. Used raw temporal splitting, segment-local feature extraction and boundary-safe context windows.
- Improved macro F1 from 0.513 to 0.967 on 2,062 eligible beats under five-fold temporal out-of-fold evaluation. These are within-record results; some abnormal beats may be synthetic and independent-session validation is pending.

Medical Imaging Labs | DSA, CT and MRI

- Implemented DSA subtraction, registration and vessel enhancement; CT Radon projections, BP/FBP, SIRT/CGLS; and MRI DICOM/NIfTI loading, spatial metadata, slice visualization and image comparisons.
- Organized three reproducible notebooks with configurable data paths, synthetic demos and numerical regression checks; verified the revised workflows on supplied coursework data.

Jarvis Research Assistant | Personal project

- Built a local paper-discovery and knowledge-management tool with multi-source retrieval, DOI/title deduplication, ranking, SQLite persistence, report export and optional Vietnamese summaries.
- Implemented a browser/CLI interface and a linked knowledge library with similarity search and clustering. Verified 91 offline checks and 16 end-to-end fixture checks.

Additional project: CT gantry ring design — structure, rotation requirements and operational safety (February 2026–Present).

TECHNICAL SKILLS

Programming and analysis: Python, PyTorch, scikit-learn, TSFEL, NumPy, pandas, SciPy, Jupyter and Git.

Medical imaging: OpenCV, scikit-image, pydicom and NiBabel.

Engineering: MATLAB, Arduino, AutoCAD, Raspberry Pi; reading component datasheets and technical documentation.

Currently learning: STM32 microcontroller architecture, peripherals and hardware fundamentals.

CERTIFICATES AND RECOGNITION

- ISAS Challenge 2025: 5th place, competing individually. OISP Scholarship: Semesters 231 and 251. First Prize, 2023 Engineering Festival, for a PPG heart-rate monitoring sensor project.

Neural Networks and Deep Learning — DeepLearning.AI | September 11, 2026

Google AI Essentials — Google | September 3, 2026 | Five-course specialization

IELTS 6.5 | MOS Excel 2019

University activities: event coordination, admission counseling and campus tours at HCMUT.